

Modular forms

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Lecture 1

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 - divisor (power) sums;
 - number of representations of an integer $m > 0$ as a sum of four squares

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- allow point counting on elliptic curves, e.g. one has for any prime p that

$$\#\{(x, y) \in \mathbb{F}_p^2 \mid y^2 - y \equiv x^3 - x^2 \pmod{p}\} = p - \varepsilon_1 + \varepsilon_5 + \varepsilon_7 - \varepsilon_{11}$$

where the ε_j are defined via some *quadratic form*

$$\varepsilon_j = \#\{(a, b, c, d) \in \mathbb{N}^4 \mid 24p = a^2 + b^2 + 11c^2 + 11d^2 \text{ and } abcd \equiv j \pmod{12}\}.$$

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In particular, **check** that we have for any $\tau \in \mathbb{H}$ that $g\tau \in \mathbb{H}$ and that

$$g_1(g_2\tau) = (g_1g_2)\tau, \qquad 1\tau = \tau, \qquad g\tau_1 = \tau_2 \Leftrightarrow \tau_1 = g^{-1}\tau_2.$$

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- 2 $\mathrm{PSL}_2(\mathbb{R}) = \overline{G} = G/\{\pm 1\}$ acts faithfully (or effectively) on $\mathbb{H}$; that is, only the elements of the subgroup $\{\pm 1\} \subset G$ act trivially on $\mathbb{H}$.

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For $g, g' \in G$ **verify** the following relation

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This is an example of a **cocycle relation**, more precisely a 1-cocycle relation.

Definition 1.7

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The group $\mathrm{SL}_2(\mathbb{Z})$ is generated by

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We will prove the proposition below.

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For any subgroup Γ of G , we define an equivalence relation on $\mathbb{H}$ via

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A **fundamental domain** $\mathcal{F}$ for the action of $\mathrm{SL}_2(\mathbb{Z})$ is given by

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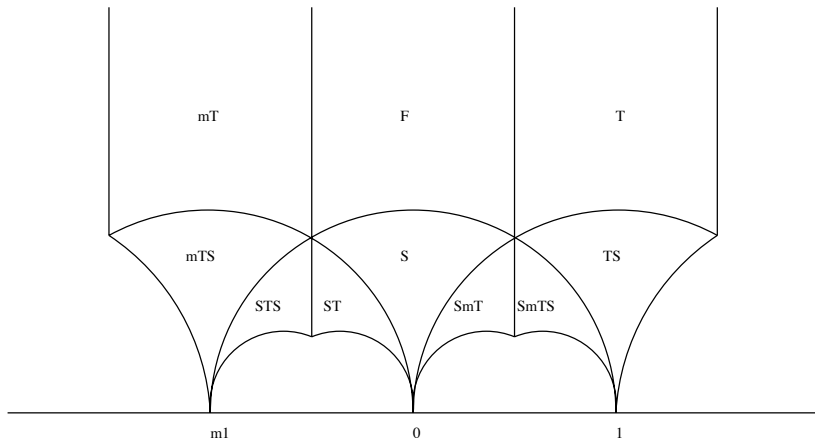
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 - for $|\operatorname{Re}(\tau)| = \frac{1}{2}$ via $\tau \mapsto \tau \pm 1$,
 - on the arc $|\tau| = 1$ via $\tau \mapsto -1/\tau$.

Figure: Fundamental Domains



Two kinds of special points: the (omitted) part of the proof yields

Corollary 1.11

For $\tau \in \mathbb{P}^1(\mathbb{C})$ we let $\Gamma_\tau = \{\gamma \in \Gamma : \gamma\tau = \tau\}$ be its *isotropy subgroup* (stabiliser in Γ).

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We have a quotient map

$$\pi : \mathbb{H} \rightarrow X$$

which makes X via the quotient topology into a topological space. (Recall that $U \subset X$ open $\Leftrightarrow \pi^{-1}(U)$ open in $\mathbb{H}$).

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Let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$. For the set of Γ -equivalence classes, we write

$$X = X_\Gamma = \Gamma \backslash \mathbb{H}.$$

We have a quotient map

$$\pi : \mathbb{H} \rightarrow X$$

which makes X via the quotient topology into a topological space. (Recall that $U \subset X$ open $\Leftrightarrow \pi^{-1}(U)$ open in $\mathbb{H}$). The map π is thus continuous, in fact open.

Two kinds of special points: the (omitted) part of the proof yields

Corollary 1.11

For $\tau \in \mathbb{P}^1(\mathbb{C})$ we let $\Gamma_\tau = \{\gamma \in \Gamma : \gamma\tau = \tau\}$ be its *isotropy subgroup* (stabiliser in Γ).

Then all points τ in $\mathcal{F}$ have trivial isotropy subgroup $\pm I$ except for the following so-called *elliptic (fixed) points*:

- ① For $\tau = i$ with stabiliser $\pm\{I, S\}$
- ② For $\tau = \omega = -\frac{1}{2} + \frac{\sqrt{-3}}{2}$ with stabiliser $\pm\{I, ST, (ST)^2\}$.
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Lemma 1.15

$\mathrm{SL}_2(\mathbb{Z})$ acts transitively on the set of rational cusps. In particular, ∞ is a representative of the single cusp of the modular group.

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